Biomedical Materials

Week 1: Introduction to Biomaterials

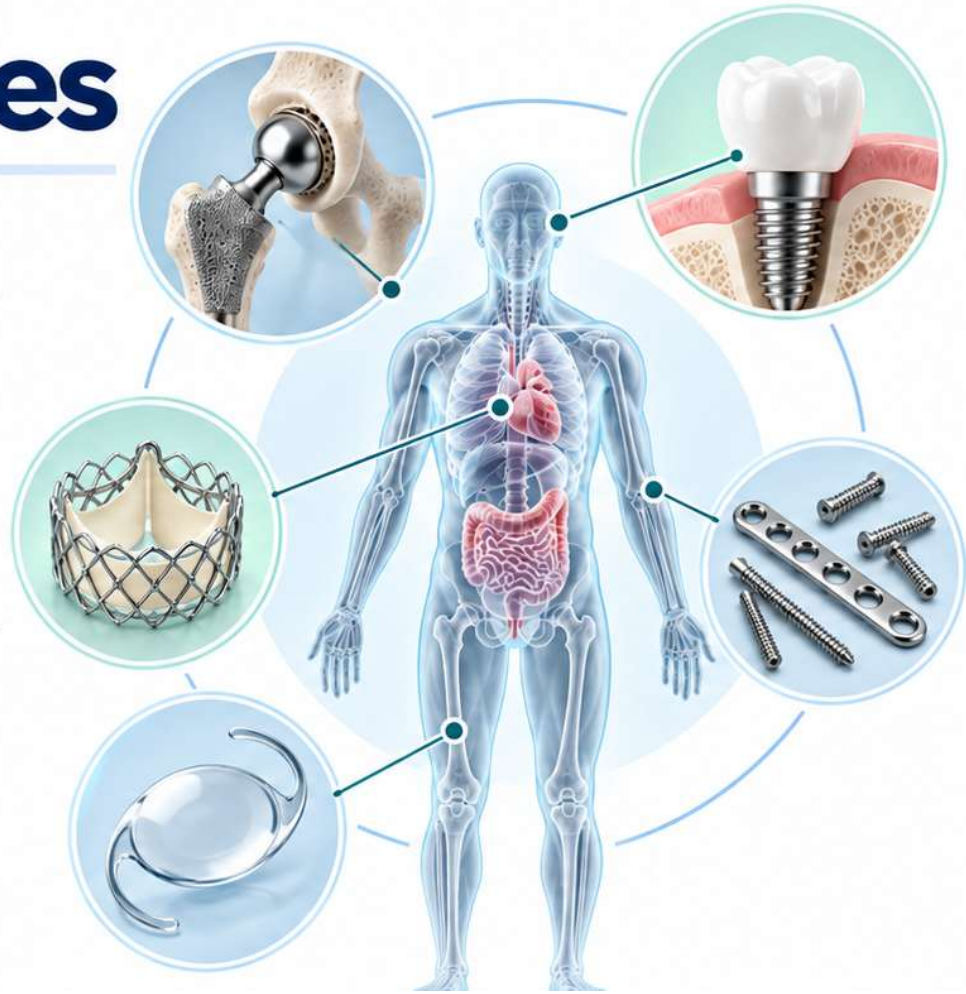
History • Definitions • Classes of biomaterials • Biocompatibility •
Biological performance • Bioinert, bioactive and biodegradable materials



Instructor:
Dr. Sumama Nuthana kalva

Learning Objectives

1. Define a biomaterial and explain its medical purpose.
2. Describe the historical development of biomaterials.
3. Identify the major classes of biomedical materials.
4. Explain biocompatibility and biological performance.
5. Distinguish between bioinert, bioactive, and biodegradable materials.
6. Relate material choice to common biomedical applications.



Lecture Roadmap

How Week 1 fits into the course

1

What are biomaterials?



2

Historical development



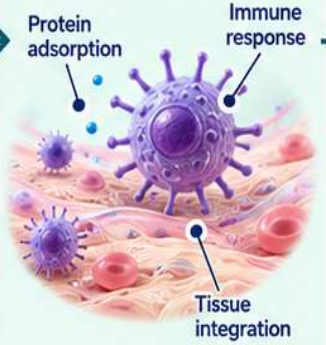
3

Major material classes



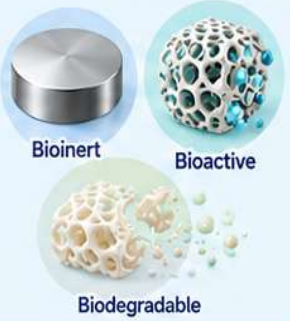
4

Biocompatibility and host response



5

Functional categories: bioinert, bioactive, biodegradable



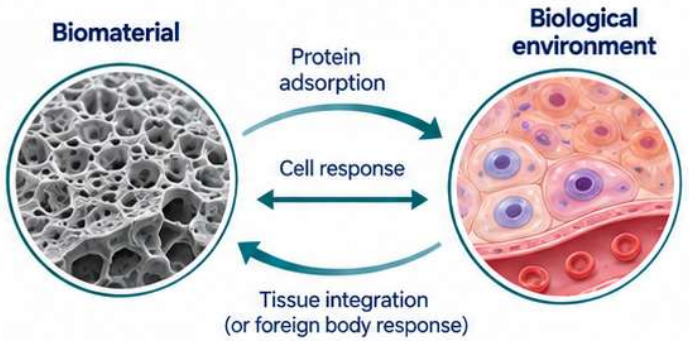
6

Key takeaways and discussion



What Is a Biomaterial?

A biomaterial is a material designed to interact with biological systems for a medical purpose such as diagnosis, treatment, repair, or replacement of tissue or function.



Common roles

- Replace damaged tissue
- Support healing or regeneration
- Restore function
- Deliver drugs or signals
- Assist diagnosis or monitoring

Examples



Hip and knee implants

Load-bearing joint replacement



Dental implants

Tooth replacement and restoration



Cardiovascular stents

Keep vessels open and restore blood flow



Contact lenses

Vision correction and drug delivery

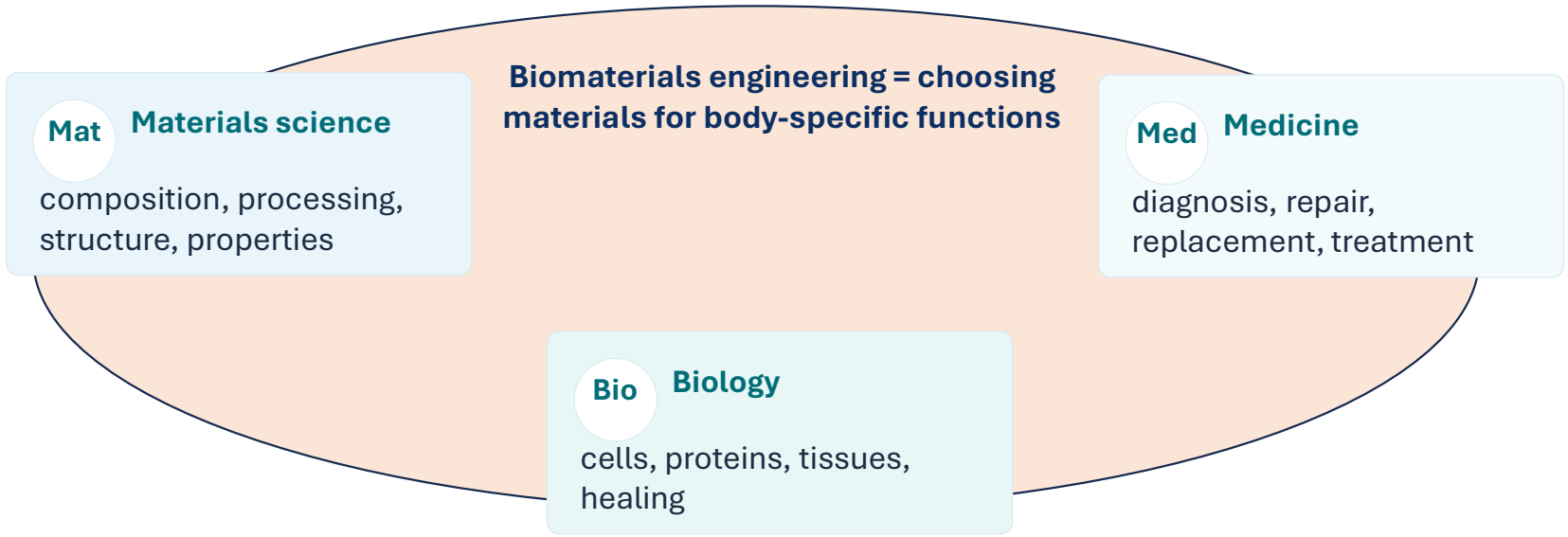


Sutures or catheters

Support healing and deliver therapies

Biomedical Materials as an Engineering Field

The course connects material structure, biological response, and clinical function



A Brief History of Biomaterials

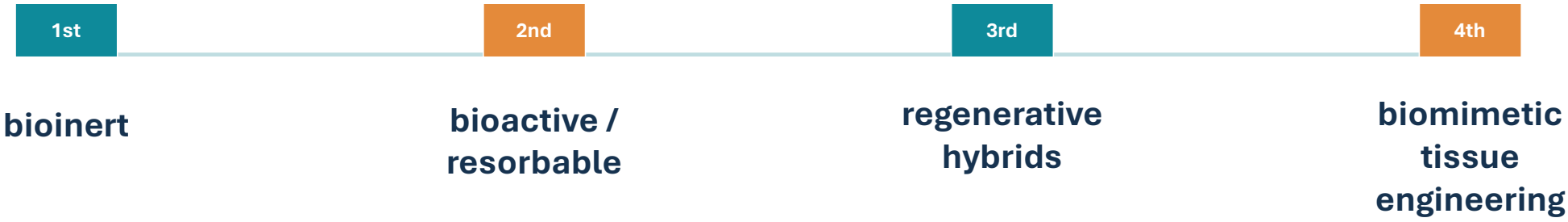


Historical Highlights related to Biomaterials

- 1504 Iron prosthetic 보철 輔綴 hand with flexible finger joints
- 1807 Development of an endoscope for minimally invasive surgery
- 1847 Introduction of silver amalgam for dental fillings
- 1888 First reports on use of contact lenses to correct vision
- 1905 Early attempt at an artificial hip replacement
- 1949 Role of immune system in tissue rejection identified
- 1951 First artificial heart valve implanted
- 1953 Development of the heart-lung machine
- 1980 First successful single-channel cochlear 달팽이관 implant in a child
- 1982 Implantation of the Jarvik-7, a permanent total artificial heart
- 1993 FDA approval of left ventricular assist device (LVAD) as a bridge to heart transplantation

Four generations of biomaterials

A teaching framework for how material goals evolved



1st generation: inert and often ad hoc

Early biomaterials were often selected from available materials

- Early use was based on materials physicians already had access to.
- Examples include gold fillings, mercury or silver, wooden teeth, glass eyes and early PMMA.
- Main goal: the material should be tolerated and not produce obvious harm.
- This stage laid the foundation for later engineering approaches.
- The limitation was lack of deliberate biological design.

Goal

Goal
minimal adverse reaction

Design style

Design style
available materials, ad hoc repair

Limitation

Limitation
little control of interface or long-term response

2nd generation: engineered implants

Physicians and engineers began designing for specific performance

- Material selection became more deliberate and property-driven.
- Examples include titanium alloys, Co–Cr–Mo implants and UHMWPE bearing surfaces.
- Heart valves and joint components became engineered systems.
- Collaboration between clinicians and engineers became more important.
- The goal shifted from simply tolerating materials to reliable device performance.

3rd generation: bioengineered and nano-enabled

Materials designed to interact more actively with the host

- Bioengineered implants combine materials at smaller length scales.
- Nanocomposites, nano oxides and nano polymers can support drug delivery and host interaction.
- Novel blends combine synthetic polymers with biopolymers or metals with polymers.
- The aim is improved interaction with the host system.
- This generation connects materials science with cell-level behavior.

Nanocomposites
Nanocomposites
small-scale control of properties

Blends
Blends
synthetic + natural polymers

Drug delivery
Drug delivery
release therapeutic agents locally

Host interaction
Host interaction
guide cells or tissue response

4th generation: biomimetic and tissue-engineered

Materials that mimic or guide biological processes

- Biomimetic materials mimic biological structure and function.
- They may act as a matrix for cell growth and signaling.
- They can deliver proteins, genes or drugs depending on design.
- The ideal outcome may be tissue regrowth rather than simple replacement.
- Scaffolds may eventually resorb as new tissue forms.

Why Biomedical Materials Matter



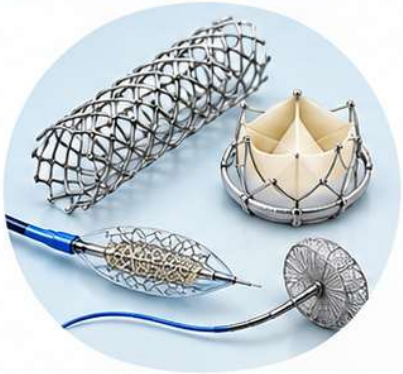
Orthopedics

Bone plates, screws, hip stems, and knee components restore mechanical support and mobility.



Dentistry

Implants, crowns, orthodontic wires, and fillings help restore oral function.



Cardiovascular

Stents, grafts, and heart-valve components support blood flow and cardiac function.



Ophthalmic / Soft Tissue

Contact lenses, intraocular lenses, catheters, and sutures support vision and soft-tissue care.

Biomedical materials improve quality of life by restoring function and supporting treatment.

Major Classes of Biomedical Materials

Metals



Strong and tough, widely used for load-bearing implants.

Examples:
stainless steel, Co-Cr alloys, titanium alloys

Ceramics



Hard, wear-resistant, and often bioactive or bioinert.

Examples:
alumina, zirconia, hydroxyapatite

Polymers



Lightweight and flexible, useful in both permanent and temporary devices.

Examples:
UHMWPE, PMMA, silicone

Composites



Combine properties of two or more materials.

Examples:
fiber-reinforced and polymer-ceramic systems

Natural / Biobased



Derived from biological sources and often used in regenerative applications.

Examples:
collagen, chitosan

Not all biomaterials are load-bearing

Metals as Biomedical Materials

Why metals are common in load-bearing devices

- High strength and toughness
- Good fatigue resistance when properly designed
- Can be machined, forged, cast, or additively manufactured
- Surface oxide layers can improve corrosion resistance
- Often used in orthopedics and dentistry

Metal

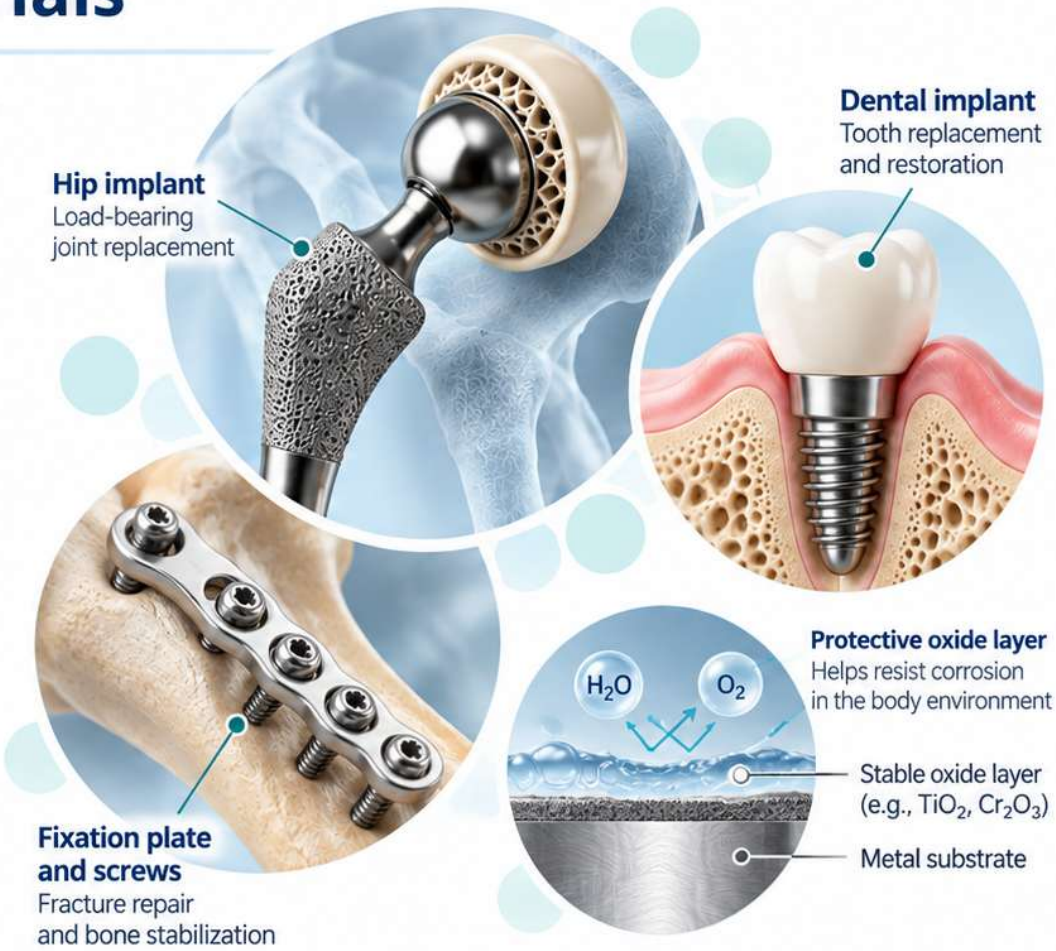
Examples

316L stainless steel
Co-Cr alloys
Titanium and Ti alloys
NiTi shape-memory alloy

Risk

Key limitation

Metals are strong, but they can still corrode, wear, release ions, or mismatch bone stiffness.



Common Metallic Biomaterials

First comparison for undergraduate students

SS 316L stainless steel



Advantages:
- Cost-effective
- Good strength
- Used in temporary fixation

Limitations:
- Corrosion and Ni sensitivity concerns

CoCr Co-Cr alloys



Advantages:
- Very wear-resistant
- High strength
- Used in joint surfaces

Limitations:
- High stiffness and ion release concerns

Ti Titanium alloys



Advantages:
- Good corrosion resistance
- Lower modulus than Co-Cr
- Osseointegration-friendly

Limitations:
- Lower wear resistance than Co-Cr

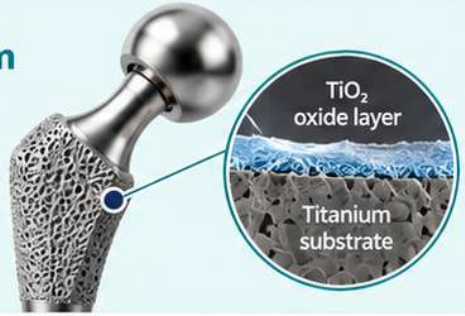
Why Titanium Is Often Preferred

A simple materials-selection example

Ox

Protective oxide film

Stable TiO_2 layer supports corrosion resistance.



ρ

Good specific strength

Strong enough for many implants with relatively low density.



Bone

Bone compatibility

Surfaces can support bone attachment and osseointegration.



AM

Processing flexibility

Can be wrought, cast, machined, or additively manufactured.



Selection is not only "strongest material" — it is compatibility with the full design problem.

Ceramic Biomaterials

Hard, wear-resistant, and sometimes bioactive

- High hardness and wear resistance
- Good chemical stability
- Can be bioinert or bioactive depending on composition
- Generally brittle compared with metals
- Used in joint surfaces, dental restorations, bone repair, and coatings

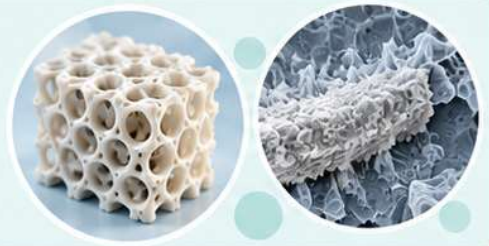
Al_2O_3 Bioinert ceramics

Alumina and zirconia are often used where low wear and chemical stability are needed.



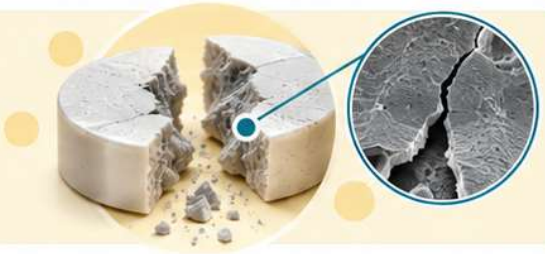
HA Bioactive ceramics

Hydroxyapatite and bioactive glass can interact positively with bone.



! Design caution

Brittleness must be considered in load-bearing designs.

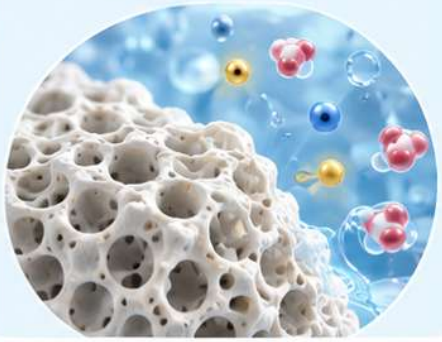


Hydroxyapatite and Bioactive Glass

Examples of bioactive ceramic behavior

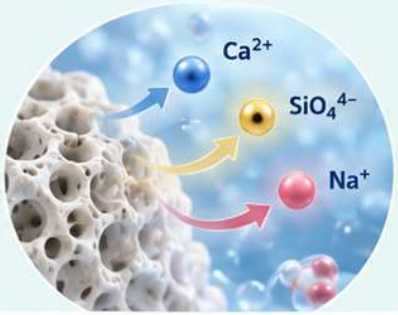
1 Material surface

HA or bioactive glass exposed to body fluid



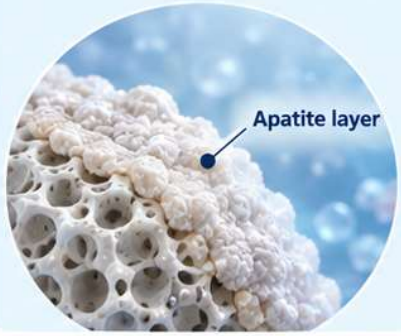
2 Ion exchange

Surface reactions and release of ionic species (e.g. Ca^{2+}, SiO_4^{4-}, Na^+) into body fluid



3 Apatite layer

A bone-like apatite layer forms on the surface in body fluid



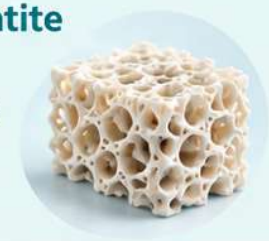
4 Tissue bonding

The apatite layer enables potential bonding to bone



HA Hydroxyapatite

Chemically similar to the mineral phase of bone; often used as coatings or bone fillers.



Glass Bioactive glass

Designed to form a bone-bonding surface reaction layer.



? Engineering question

How fast should the surface react for the intended application?



Polymeric Biomaterials

Flexible, processable, and versatile

- Lower density than metals and ceramics
- Wide range of stiffness, flexibility, and degradation behavior
- Can be shaped by molding, extrusion, spinning, printing, or casting
- Useful for soft devices, coatings, drug delivery, and temporary implants
- May absorb water, age, degrade, or wear depending on polymer chemistry

PE

Permanent examples

UHMWPE bearing liners, silicone soft devices, PMMA bone cement



PLA

Temporary examples

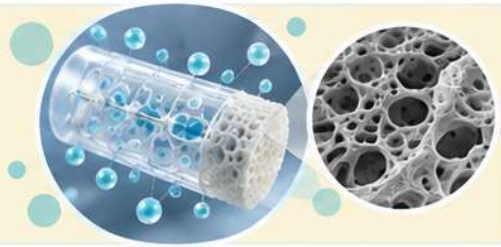
PLA/PGA sutures, absorbable fixation, drug-release matrices



t

Key idea

Polymer performance is strongly time- and environment-dependent.



Polymer Examples in Devices

Linking polymer properties to function

UHM **UHMWPE**

low friction bearing surfaces in hip/knee prostheses



PMM **PMMA**

bone cement and some dental/ocular applications



Sil **Silicone**

flexible soft-tissue devices and catheters



PLA **PLA/PGA**

absorbable sutures and biodegradable scaffolds



Polymer selection often balances flexibility, degradation, wear, and sterilization compatibility.

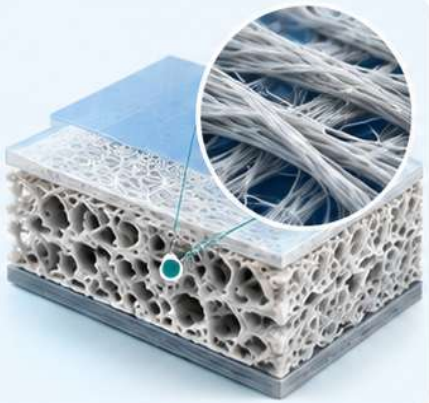
Composites and Natural Materials

Combining or borrowing from biological structures

C+ Composites

Composites combine two or more material phases to obtain properties that a single phase cannot provide.

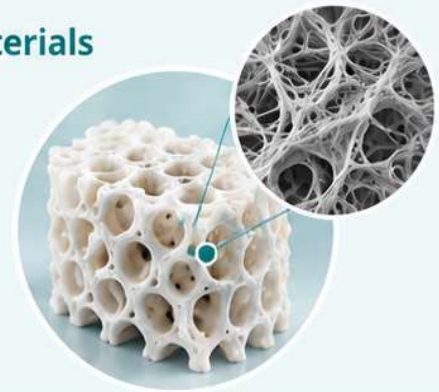
Examples: fiber-reinforced polymers, polymer–ceramic bone substitutes.



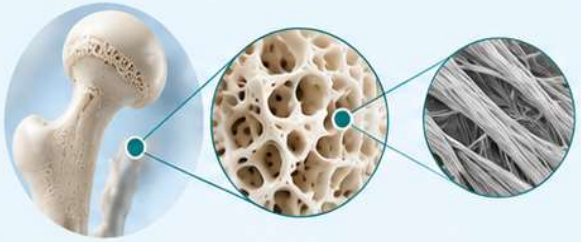
Bio Natural / biobased materials

Materials derived from biological sources can support regenerative applications.

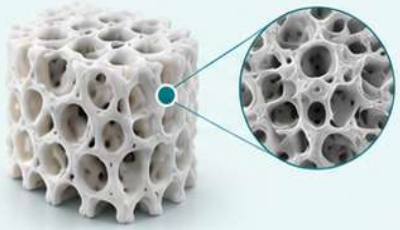
Examples: collagen, chitosan, alginate, decellularized matrices.



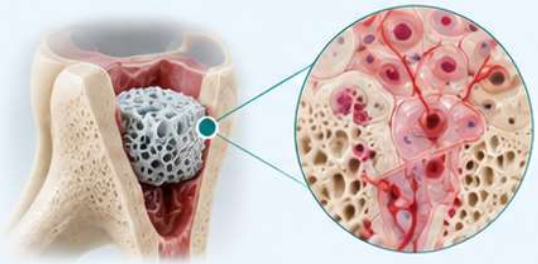
Natural tissue idea
hierarchical structure



Material design
composite or scaffold



Biological goal
guide healing



Properties that matter

1 Mechanical



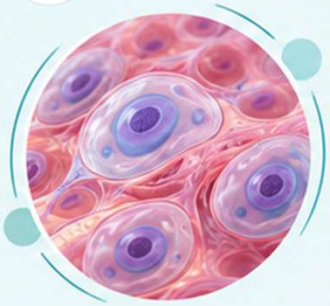
strength, stiffness, fatigue, wear

2 Chemical



corrosion, degradation, water uptake

3 Biological



biocompatibility, immune response, infection

4 Processing



manufacturing, surface finishing, sterilization

Mechanical Properties of Biomaterials

Properties must match the device function and loading environment

Strength
Resists applied force.

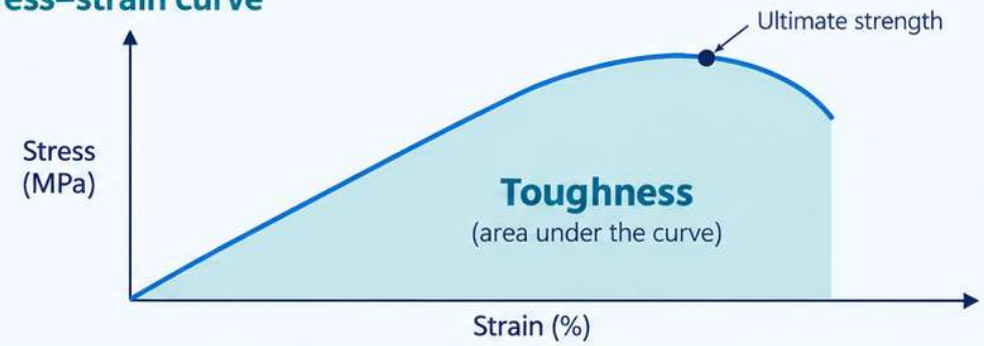
Toughness
Absorbs energy before fracture.

Elastic modulus
Controls stiffness and load transfer.

Wear resistance
Limits debris generation.

Hardness
Resists surface indentation and scratching.

Stress–strain curve



Wear, debris generation, and biological response



Two materials rubbing in a biological environment

Wear debris
(metal and/or polymer particles)

Possible inflammatory response



Metal debris may release ions e.g., Co^{2+}, Cr^{3+}, Ni^{2+}.

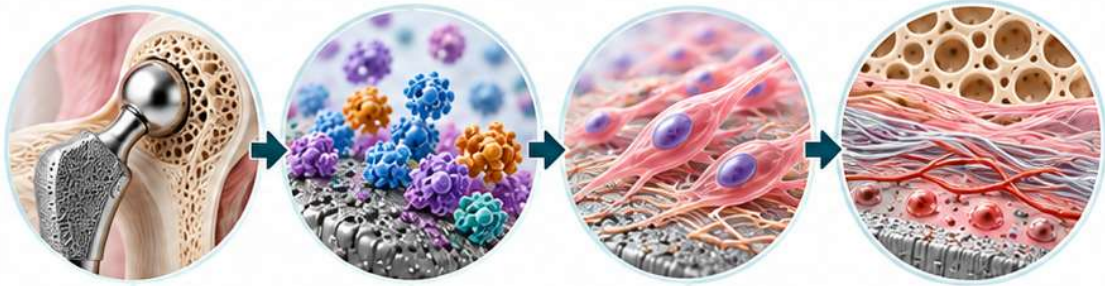
Polymer debris can trigger foreign body / chronic inflammatory reactions.

Biocompatibility and Biological Performance

Biocompatibility

- Ability of a material to perform its intended function in the body without causing unacceptable harm.
- Depends on material composition, surface, shape, and location in the body.
- The best material is not always the strongest; it must also be biologically acceptable.
- Biocompatibility is application-specific.

Biological performance



Implant placed → Protein adsorption → Cell and tissue response → Outcome: integration, encapsulation, or failure

Key idea: material success depends on both engineering performance and host response.

Biocompatibility

A material must function without unacceptable harm

- Biocompatibility is not a single universal material property.
- It depends on material composition, surface state, shape, device location, and time in the body.
- A material can be acceptable in one application but unsuitable in another.
- Testing must consider the intended use and tissue contact.

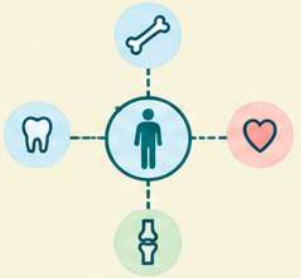
Ti Example

Titanium may be suitable for bone-contacting implants but not automatically ideal for every device.



Important concept

Biocompatibility is application-specific.



Not enough
"This material is biocompatible."

Better
"This material is biocompatible for this device, in this location, for this duration."

The Body Is a Challenging Environment

Why ordinary materials may fail in biomedical use

H_2O Water + ions

electrolyte-like fluid can drive corrosion and swelling



Pr Proteins

rapid adsorption changes the surface that cells see



Cell Cells

immune and tissue cells react to surface chemistry and topography



Load Mechanical loads

cyclic loading, sliding, and wear can occur



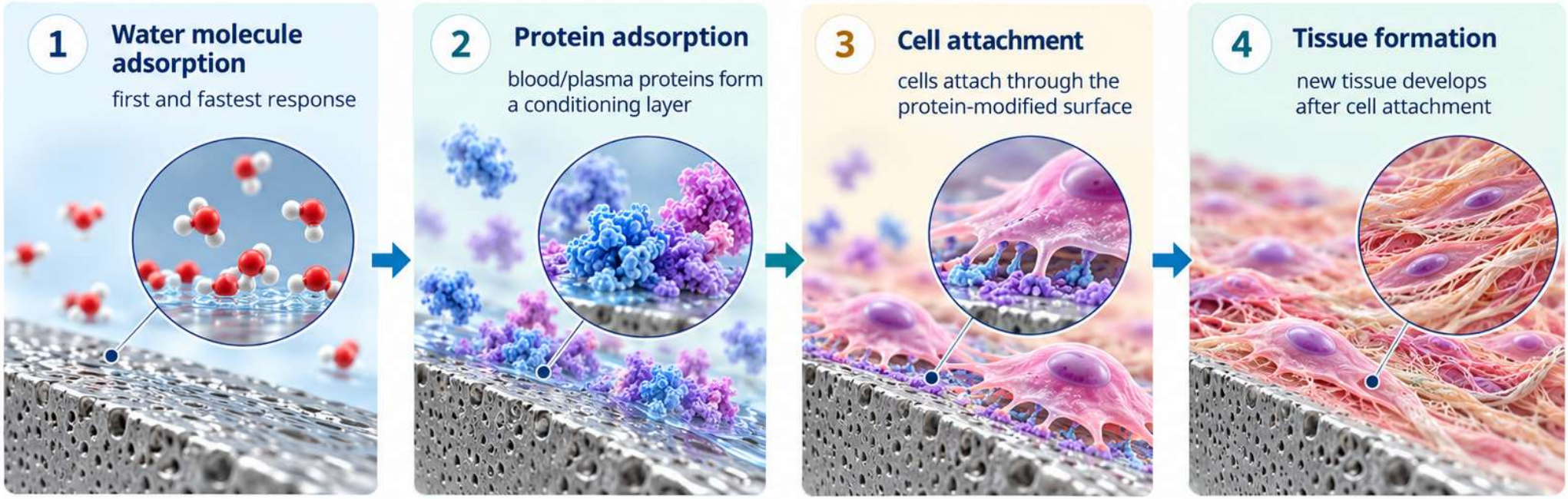
Bug Microbes

infection risk requires sterilization and surface control



Tissue–Biomaterial Interaction

Biological response begins immediately at the surface



✓ **The early adsorbed layer translates material properties into biological response:** biocompatibility, inflammation, or immune response.

Inflammation and Foreign Body Response

A simplified sequence for introductory teaching

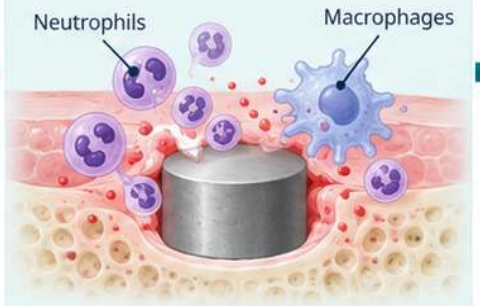
Injury + Implant

surgery and material placement



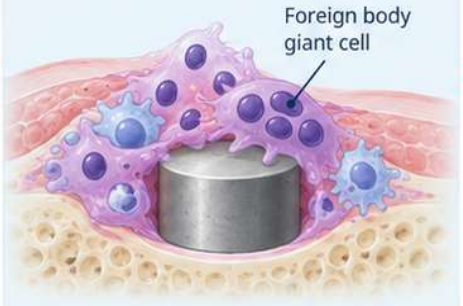
Acute inflammation

early immune response and cleaning



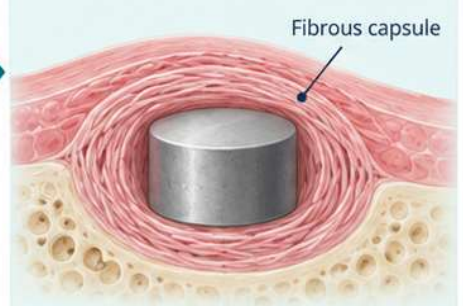
Chronic response

longer-term immune activity if unresolved



Fibrous capsule

possible isolation of the material



! Why this matters

Excessive or prolonged inflammation can compromise device function.



Ctrl Engineering strategy

Control surface, geometry, cleanliness, motion, and material degradation.



Host response can be local or systemic

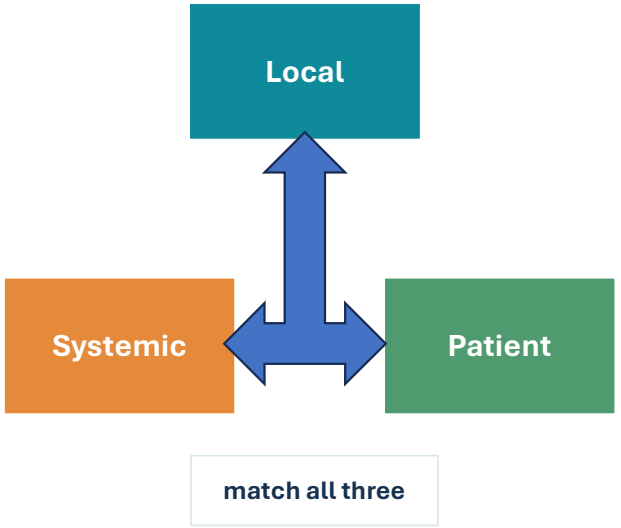
Both responses must be considered during design

Local response occurs near the material or device surface.

Systemic response affects the whole body or distant tissues.

Wear particles, ions and degradation products can trigger reactions.

Patient-to-patient variability means one person may tolerate a material that affects another.



Possible Biological Outcomes

Integration, encapsulation, degradation, or failure

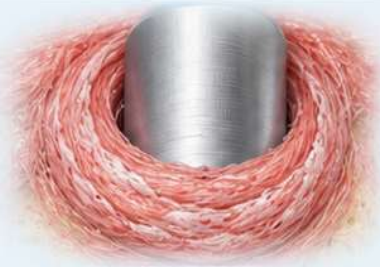
✓ Integration

desired tissue bonding or stable fixation



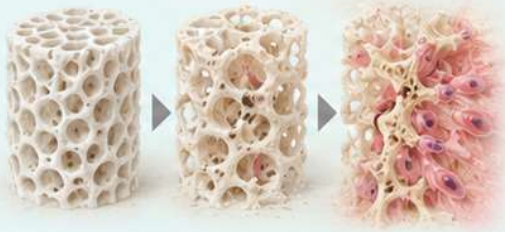
Caps Encapsulation

fibrous tissue isolates device from host



↓ Resorption

biodegradable material is replaced or cleared



× Failure

infection, loosening, wear, fracture, thrombosis, toxicity



Biological performance is judged against the intended function and lifetime.

Examples of host response concerns

The response may come from the material, surface or degradation products

Inflammation near a cardiovascular stent or catheter.

Local acidity around a degrading polymer such as PLA.

Metal ion sensitivity from nickel, chromium or cobalt-containing alloys.

Protein adsorption and cell attachment that redirect tissue response.

Bacterial attachment leading to biofilm and infection.

The material–tissue interface

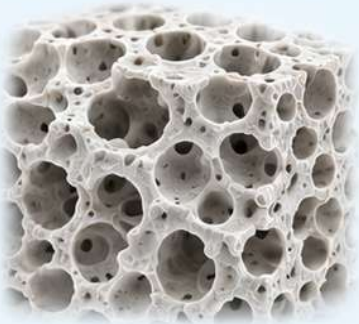


Surface Properties of Biomaterials

The surface often controls early biological response

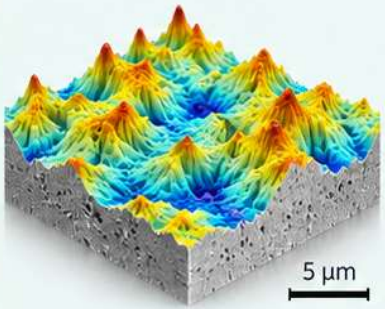
1 Porosity

Voids and cavities accessible to molecules.



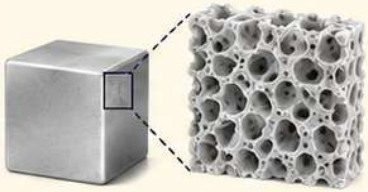
2 Surface roughness

Topography at micro- and nano-scale.



3 Surface area

Exposed surface available for interaction.



Smooth surface (lower area)

Rough/porous surface (higher area)

4 Wettability

Tendency of fluid to spread or adhere.

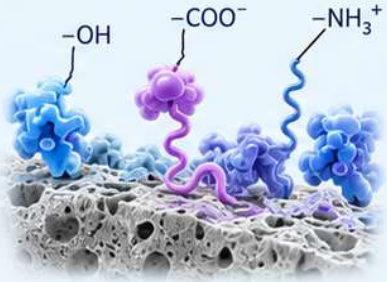


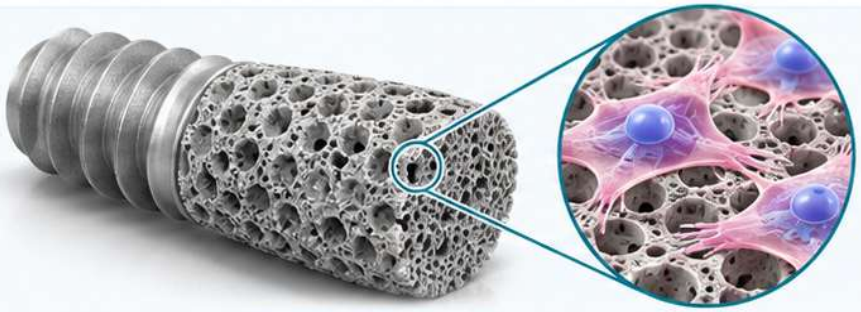
More hydrophobic (higher contact angle)

More hydrophilic (lower contact angle)

5 Surface chemical structure

Functional groups and charge.





These properties affect **protein adsorption, osteoblast proliferation, and tissue reaction.**

Surface modification: making the interface work

The surface controls the first contact with the body

Surface treatment aims to improve compatibility and reduce harmful reactions.

Physical or mechanical methods can change roughness and texture.

Chemical methods can change functional groups, charge or hydrophilicity.

Biological methods can immobilize cells, proteins, peptides or growth factors.

Surface design influences cell attachment, bone bonding and infection control.

Sterility and Infection: Introductory View

A biomaterial must be safe before it reaches the patient

- Sterilization removes or inactivates microorganisms before device use.
- Methods must not damage the material or change its function.
- Bacterial adhesion and biofilm formation can cause serious implant failure.
- Surface design and surgical handling both influence infection risk.



Ster **Common sterilization routes**

steam, ethylene oxide, radiation, plasma-based methods



Steam (autoclave) | Ethylene oxide | Radiation | Plasma-based methods

Poly **Materials issue**

Some polymers degrade or age after heat or radiation exposure.



Bioinert, Bioactive, and Biodegradable Materials

Bioinert

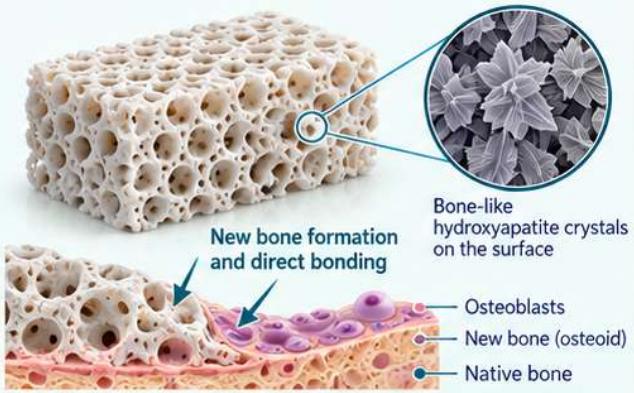
Causes minimal interaction with surrounding tissue.



Examples:
alumina, zirconia, titanium alloys, medical-grade stainless steel

Bioactive

Interacts positively with tissue and may bond to bone or stimulate a response.



Examples:
hydroxyapatite, bioactive glass, bioactive ceramics

Biodegradable

Gradually breaks down in the body over time.



Initial scaffold
Provides mechanical support

Progressive degradation
Maintains structural integrity while tissue regenerates

Resorbed
Replaced by new tissue

Examples:
PLA, PGA, PLGA, magnesium-based materials, biodegradable polymers

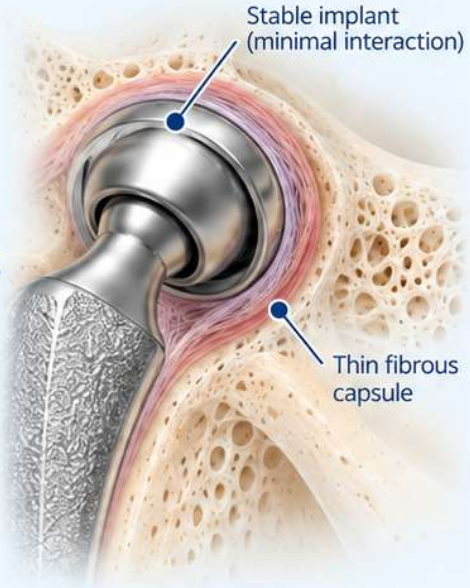
Material choice depends on the application and the desired interaction with the body.

Bioinert and Biodegradable Materials

Different design intentions for material–body interaction

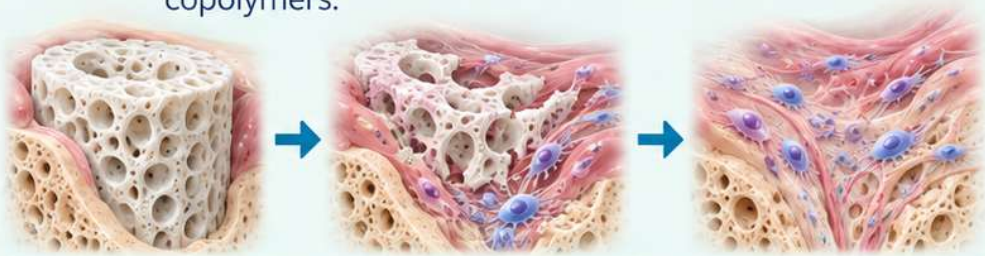
1 Bioinert

- Shows minimal interaction with surrounding tissues.
- **Examples:** stainless steel, titanium, alumina, partially stabilized zirconia, UHMWPE.
- A fibrous capsule may form around bioinert implants.



2 Biodegradable / bioresorbable

- Begins to dissolve or resorb in the body and is gradually replaced by advancing tissue.
- **Examples:** tricalcium phosphate $Ca_3(PO_4)_2$ and PLA/PGA copolymers.



Initial scaffold (porous structure)

Gradual resorption and tissue ingrowth

New tissue replaces resorbed material

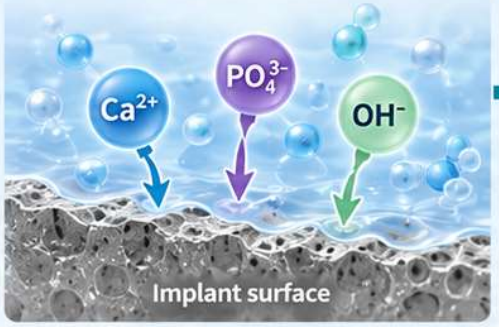
Do we want stability, resorption, or tissue replacement?

Bioactive Materials

Surface reaction can create a bond with tissue

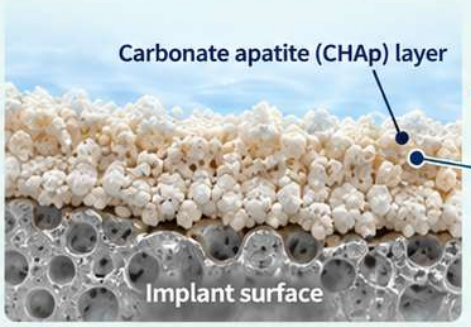
1 Ion exchange

Body fluid interacts with implant surface.



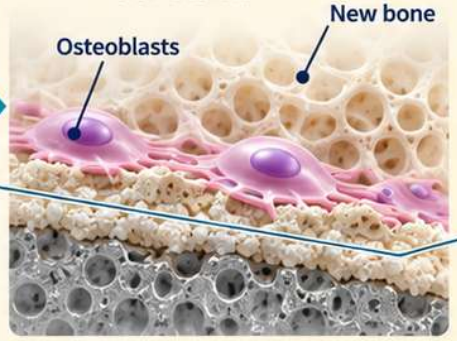
2 Carbonate apatite layer

A biologically active CHAp layer forms on the surface.



3 Tissue bonding

Bone tissue bonds to the biomaterial.



✓ **The carbonate apatite layer is chemically and crystallographically similar to the mineral phase in bone.**



Examples of bioactive materials



Bioactive glass

Silicate-based glasses that form an apatite layer in body fluid.



Bioceramics

Calcium phosphate ceramics (e.g., hydroxyapatite, tricalcium phosphate).



Bioceramic glass

Glass-ceramic materials combining glass reactivity and crystalline phases.



Composites

Bioactive particles or glass in polymer or ceramic matrices.

✓ **Bioactivity promotes tissue regeneration or specific cell responses.**

Deterioration and failure modes

Not all degradation is intentional

Metals may corrode and release ions.

Polymers may degrade, swell, absorb proteins or leach additives.

Materials may calcify, oxidize, wear or fatigue under load.

Bacterial growth and body fluids can accelerate deterioration.

Failure often results from combined chemical, mechanical and biological effects.

Chemical
Chemical
corrosion, oxidation, leaching

Mechanical
Mechanical
wear, fatigue, fracture, loosening

Biological
Biological
enzymes, bacteria, inflammation

Combined
Combined
fretting corrosion, infected wear debris

Introductory material-selection criteria

Move from vocabulary to engineering decision-making.

Mechanical	Chemical	Biological	Processing	Economics

How do we select a biomaterial?

Selection begins with the problem, not with a favorite material

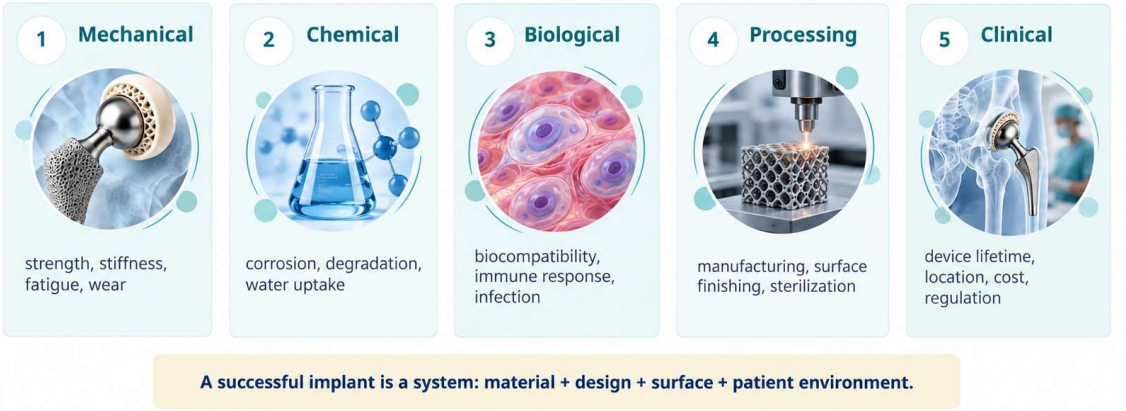
Define the function and load level.

Identify the body site and local environment.

Estimate residence time and degradation requirements.

Consider biological response, toxicity, allergy and infection.

Evaluate processing, sterilization, reproducibility and cost.



Which criterion would dominate for a hip stem? Which would dominate for a contact lens?

Function is the starting point

What biomaterials are expected to do inside or near the body

R **Replace**

damaged tissue or missing structure

S **Support**

healing, fixation, or regeneration

F **Restore**

mechanical, optical, or electrical function

D **Deliver**

drugs, signals, or therapeutic agents

M **Monitor**

diagnostic information or physiological state

Design begins with the desired function and the local biological environment.

The design triad

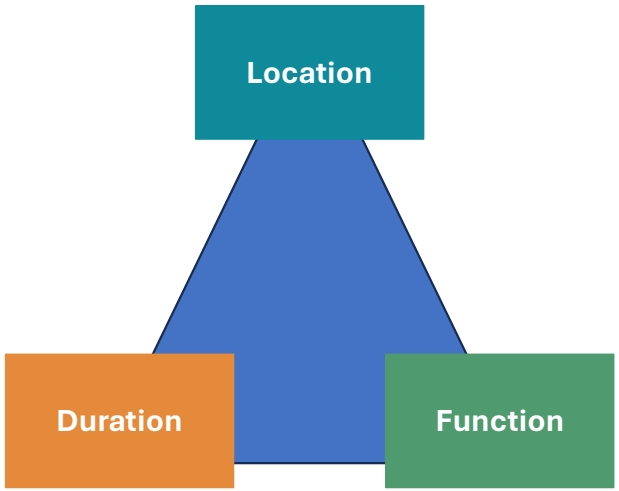
The same material can be good or poor depending on these three variables

Location determines pH, fluid chemistry, cells, proteins, salts and bacteria.

Duration determines exposure time, degradation risk and testing burden.

Function determines loads, motion, transport, sensing or tissue interaction.

Material selection changes whenever one part of the triad changes.



If a device moves from bone to blood, which part of the design triad changes most strongly?

Contact time changes design requirements

Temporary and permanent devices face very different risks

Hours: urinary catheter or temporary drainage device.

Weeks: ureteral stent or short-term wound support.

Years: cardiovascular stent or implanted sensor.

Lifetime: hip or knee replacement.

Longer residence time increases concern for wear, degradation, infection and failure.



Body environments are not equivalent

Material performance depends on local chemistry and biology

Bone sites experience mechanical loads and remodeling.

Blood-contacting devices face proteins, cells, clotting and flow.

Urinary devices face salts, bacteria and precipitation risk.

Ocular devices require oxygen permeability, comfort and optical clarity.

Skin and soft tissues bring infection, motion and healing requirements.

Mechanical properties

Important for load-bearing and moving devices

Strength determines whether a device can carry load without failure.

Elastic modulus affects load sharing with surrounding tissue.

Fatigue resistance matters under repeated cycles such as walking or valve opening.

Fracture toughness is important when cracks or flaws may exist.

Wear resistance matters at articulating surfaces.

Strength
Strength
resist fracture under load

Modulus
Modulus
match tissue and control stress shielding

Fatigue
Fatigue
survive repeated cycles

Wear
Wear
reduce debris and surface damage

Chemical and biochemical properties

The body is a reactive aqueous environment

- Corrosion resistance is critical for metallic implants.
- Hydrolysis and oxidation can degrade polymers.
- Solubility and ion release matter for ceramics and glasses.
- Surface chemistry determines protein adsorption and cell response.
- Degradation products must not cause unacceptable local or systemic harm.

Biological properties and Compatibility

A material must work with the host system

- Cytotoxicity tests examine whether cells are harmed.
- Sensitization and allergy are important for some metal ions and leachables.
- Blood-contacting materials must consider clotting and hemolysis.
- Inflammation and immune response influence long-term success.
- Biological testing must match the device application.



Processing, fabrication and sterilization

A good material must also become a reliable device

- The material must be shaped into the correct geometry and surface finish.
- Processing should produce reproducible properties at scale.
- Sterilization must not damage the device or change its performance.
- Packaging must maintain sterility until use.
- Manufacturing constraints can eliminate otherwise attractive materials.



A simple selection checklist

Students can use this checklist for any biomedical device example

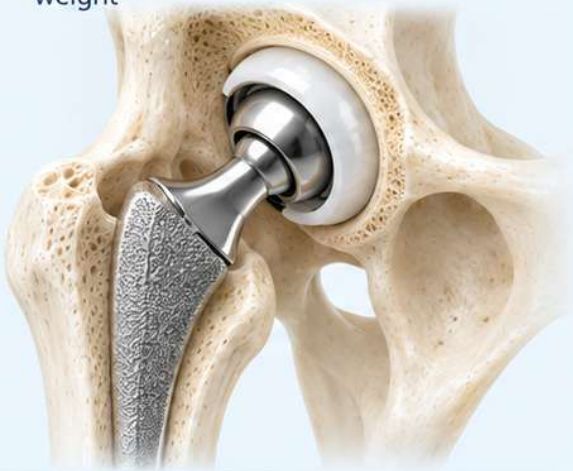
- What function must the device perform?
- Where will it be located and what will it contact?
- How long will it remain in the body?
- Which mechanical, chemical and biological properties are essential?
- How will it be fabricated, sterilized, tested and monitored for failure?

Case Study: Orthopedic Hip Implant

Material selection for load-bearing and bone contact

Hip Device function

restore joint motion and transfer body weight



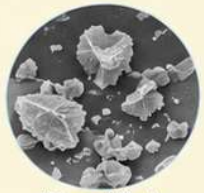
Mat Material choices

Ti alloy stems, Co-Cr or ceramic heads, UHMWPE liners



Risk Design concerns

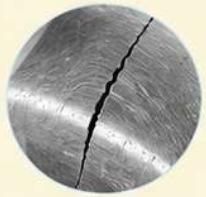
wear debris, loosening, corrosion, stress shielding, fatigue



Wear debris



Corrosion



Fatigue



Loosening



Case Study: Dental Materials

A demanding chemical and mechanical environment

- Oral environment contains saliva, ions, food chemistry, microbes, and cyclic forces.
- Dental implants often use titanium for osseointegration.
- Dental ceramics provide aesthetics, hardness, and wear resistance.
- Orthodontic wires require elasticity, corrosion resistance, and formability.



Ti **Implants**

Ti or Ti alloys + engineered surfaces



Cer **Restorations**

ceramics, composites, amalgams, polymers



? **Question**

How should aesthetics change material choice?



Case Study: Cardiovascular Devices

Blood-contacting materials have special requirements

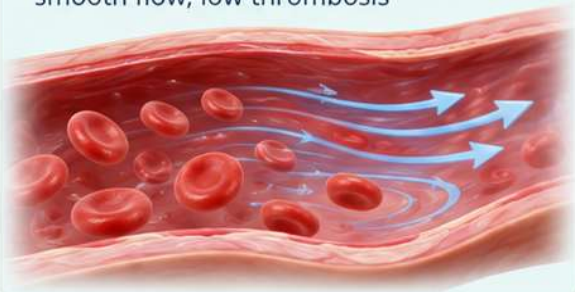
Main examples

stents, vascular grafts, heart valves, catheters



Flow Design requirements

hemocompatibility, fatigue resistance, smooth flow, low thrombosis



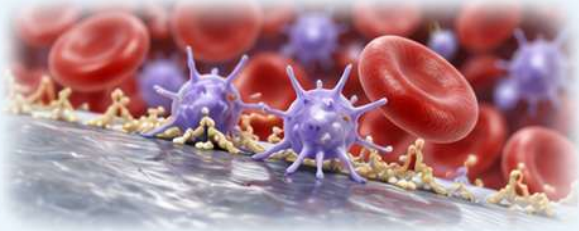
Mat Materials

metals, polymers, pyrolytic carbon, biological tissues



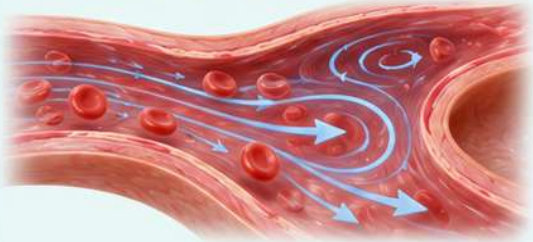
Blood contact

proteins and platelets interact with surface



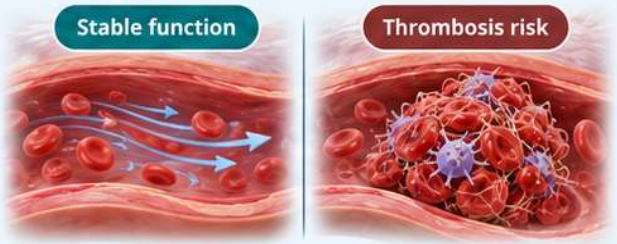
Flow environment

geometry controls turbulence and shear



Outcome

stable function or thrombosis risk



Case Study: Ophthalmic and Soft-Tissue Materials

When transparency and flexibility become design requirements

Lens **Contact lenses**

transparent, oxygen-permeable, wettable, comfortable



IOL **Intraocular lenses**

optically clear, stable, implantable polymers



Cath **Catheters**

flexible polymers with controlled surface friction



Sut **Sutures**

temporary mechanical support with biocompatible response



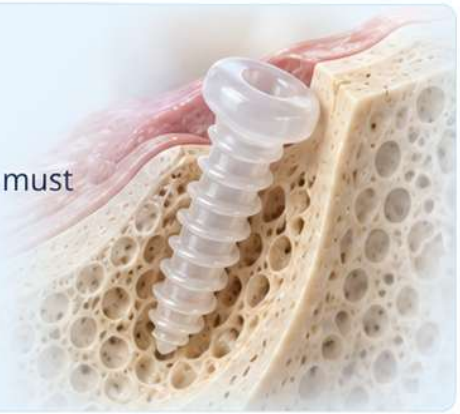
Soft-tissue applications often prioritize flexibility, surface comfort, and infection control.

Class Activity: Select a Material

Apply the Week 1 design checklist

A Scenario A

Temporary bone fixation screw: must hold during healing and then gradually disappear.



B Scenario B

Permanent hip bearing surface: must resist wear under repeated motion for many years.



- Which material class would you start with?
- What properties are most important?
- What biological response is desired?
- What could cause failure?

Talk Discussion output

Students should justify material choice using function, environment, and host response.



Biomedical Materials

Week 1: Introduction to Biomaterials

History • Definitions • Classes of biomaterials • Biocompatibility • Biological performance • Bioinert, bioactive and biodegradable materials



Instructor:
Dr. Sumama Nuthana kalva